

UTKLIPPAN, Sweden

WMO: 026320

Lat: 55.953N

Lon: 15.703E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.9	24.2	15.1	11.8	24.7	18.1	13.7	27.5	39.1	38.0	36.8	36.6	17.2	20	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	73.4	67.7	71.3	66.6	69.6	65.4	69.3	72.2	67.4	70.3	65.9	68.8	10.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	103.6	71.5	66.2	96.4	69.3	64.6	91.1	67.9	33.5	72.1	31.9	70.6	30.7	68.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.0	31.9	29.2	19.6	74.6	4.9	2.8	16.1	76.6	13.2	78.2	10.4	79.8	6.9	81.8
			19.4	69.4	4.4	3.3	16.3	71.7	13.7	73.6	11.2	75.4	8.0	77.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	34.3	34.2	36.1	41.4	49.9	58.5	63.7	64.5	57.9	50.3	43.8	38.3	
	DBStd	11.68	4.77	4.82	4.20	3.05	4.06	3.75	4.29	3.70	4.10	4.60	4.50	5.69	
	HDD50	2276	488	442	430	258	52	0	0	0	1	53	190	362	
	HDD65	6343	953	862	895	708	467	196	77	52	214	455	637	827	
	CDD50	1486	0	0	0	0	50	256	425	450	239	63	3	0	
	CDD65	79	0	0	0	0	0	2	37	37	3	0	0	0	
	CDH74	51	0	0	0	0	0	0	30	21	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	15.7	18.2	17.8	16.0	13.9	13.4	13.1	12.6	13.8	15.6	17.2	18.2	19.2	
Precipitation	PrecAvg	22.5	1.7	1.5	1.3	1.1	1.5	1.7	1.9	2.5	2.4	2.4	2.4	2.1	
	PrecMax	28.2	3.4	3.0	3.0	2.1	4.2	4.2	5.9	7.9	6.1	6.1	6.3	4.9	
	PrecMin	16.7	0.1	0.3	0.2	0.0	0.4	0.0	0.0	0.5	0.8	0.5	1.1	0.9	
	PrecStd	3.3	0.9	0.6	0.7	0.6	0.9	1.0	1.2	1.7	1.3	1.3	1.2	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.7	43.4	45.8	51.4	61.6	70.6	77.1	76.7	70.2	60.6	54.5	48.9	
		MCWB	42.2	40.9	42.2	46.5	56.8	64.3	70.5	70.4	64.0	58.1	52.7	47.3	
	2%	DB	42.5	42.2	43.4	48.6	59.5	67.6	74.3	72.9	67.4	58.6	52.5	47.7	
		MCWB	41.2	40.2	40.7	44.8	55.2	62.7	68.9	66.9	62.2	56.6	51.0	46.1	
	5%	DB	41.4	41.2	42.1	47.1	57.6	65.7	72.0	71.2	65.2	57.4	50.8	46.5	
		MCWB	40.2	39.7	40.0	43.9	53.6	61.3	67.5	66.3	61.0	55.4	49.3	45.2	
	10%	DB	40.5	40.2	41.0	45.5	55.8	63.9	70.1	69.7	63.3	56.2	49.0	45.1	
		MCWB	39.3	38.9	39.2	42.9	52.4	60.2	66.2	65.4	59.8	54.4	47.5	43.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	42.0	42.9	47.4	58.1	65.8	72.0	71.1	65.6	59.0	53.6	47.8
			MCDB	43.4	42.7	44.8	49.8	60.7	68.3	75.2	74.0	68.4	59.8	54.4	48.3
2%		WB	41.5	40.8	41.4	45.6	55.9	63.6	70.0	69.0	63.5	57.2	51.3	46.6	
		MCDB	42.1	41.7	42.8	47.9	58.7	66.4	73.0	71.9	66.1	58.1	52.2	47.5	
5%		WB	40.6	40.0	40.4	44.4	54.3	62.1	68.3	67.0	61.9	55.8	49.6	45.5	
		MCDB	41.3	41.0	41.6	46.6	57.0	64.8	71.2	69.9	64.2	56.9	50.7	46.4	
10%		WB	39.6	39.2	39.6	43.2	52.8	60.8	66.6	65.9	60.3	54.7	47.9	44.1	
		MCDB	40.4	40.3	40.8	45.2	55.2	63.4	69.8	68.7	62.6	55.9	48.7	45.0	
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.3	5.1	5.8	6.3	7.1	6.9	6.4	5.8	4.7	4.2	4.3	
		MCDBR	4.1	4.3	6.3	7.8	7.9	8.8	8.6	7.2	7.1	5.3	4.2	3.6	
		MCWBR	4.3	3.7	4.8	5.2	5.1	5.6	5.1	5.0	4.9	4.4	4.6	3.8	
	5% WB	MCDBR	4.1	4.2	5.8	7.3	7.3	7.9	7.5	6.3	6.0	4.5	4.1	3.5	
		MCWBR	4.6	3.8	4.7	5.1	5.1	5.4	4.9	4.9	4.9	4.3	4.6	3.9	
Clear-Sky Solar Irradiance	taub	0.315	0.341	0.359	0.391	0.389	0.380	0.406	0.405	0.375	0.361	0.343	0.308		
	taud	2.294	2.314	2.322	2.303	2.380	2.438	2.386	2.385	2.444	2.390	2.310	2.262		
	Ebn at Noon	186	227	254	262	271	274	264	255	248	218	172	156		
	Edh at Noon	17	24	30	36	35	34	35	33	27	23	17	14		
All-Sky Solar Radiation	RadAvg	130	325	784	1348	1778	1905	1731	1398	971	457	174	93		
	RadStd	24	59	75	149	169	119	191	116	105	77	32	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-279	+108	N/A

Nomenclature: See separate page